

Learning-Aided Privacy-Utility-Oriented Adaptive Rate Vector Quantization

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Abstract—This work presents *Privacy-Utility Rate Adaptive Vector Quantization (PURAveQ)*, a deep learning-based vector quantization framework for remote inference that achieves multi-rate, privacy-aware, and task-oriented compression. PURAveQ combines learned representation with adaptive vector codebooks, alongside mutual-information-driven training mechanism that balances utility and privacy while enforcing nested codebooks to support multiple quantization rates within a single model. Experimental results demonstrate that the learned quantizer effectively conceals private attributes without compromising inference accuracy, enabling reliable and adaptable remote inference across dynamic communication conditions.

I. INTRODUCTION

Remote inference systems are becoming increasingly central in various technologies [1], enabling data collected at edge devices to be processed for inference at remote servers [2]. Operation relies on conveying information extracted from high-dimensional sensory data over bandwidth- and energy-limited links [3], [4]. Remote inference mechanisms must therefore perform *efficient* feature compression that minimizes latency, while also being *adaptive* to dynamic communication conditions and *privacy-preserving*, ensuring that transmitted representations reveal only the task-relevant information [5]. Jointly achieving efficiency, adaptivity, and privacy, requires learned representations that support variable-rate communication and are optimized not merely for reconstruction, but for the desired downstream task.

Existing research on remote inference and task-oriented compression can be broadly categorized into model-based and data-driven approaches. Model-based methods treat source coding and inference as separate stages [6], with schemes exploiting task-awareness proposed in quantization for specific tasks viewed as linear [7], [8] or quadratic functions [9], [10], and for semantics-aware coding [11]. Data-driven schemes leverage deep neural networks (DNNs) to jointly learn feature extraction, compression, and inference mappings [12], [13], using quantization layers such as uniform [14], scalar [15], [16], or vector quantization [17], [18], that can be extended to support variable rates [19] and successive refinement at the decoder [20]. While these techniques effectively tailor the compressed representation to the inference task, they typically disregard the possibility that transmitted features may inadvertently expose sensitive information.

Privacy preservation is studied under several paradigms. A leading framework is that of *differential privacy (DP)*, which provides formal guarantees on concealing the contribution of individual data samples through randomized perturbations [21]. DP mechanisms are often realized by injecting controlled noise, and can be combined with quantization [22]–[24]. However, such methods are designed for dataset-level privacy and typically induce notable distortions, rendering them less suitable for

remote inference, where privacy concerns arise at the level of each transmitted sample. Another candidate paradigm is *homomorphic encryption*, which enables computation directly in the encrypted domain [25], [26], but comes at the cost of substantial computational and communication overhead, often incompatible with latency-sensitive edge inference. A third line of work adopts *information-theoretic* privacy measures, formulating privacy as the minimization of Mutual information (MI) between the representation and the private attributes [27]. While classical approaches such as the privacy funnel [28] are limited to models where MI can be computed analytically or through heuristic optimization [29], recent advances in deep learning have facilitated accurate estimation and optimization of MI with high-dimensional data [30] and latent representations [31]–[33]. Still, its potential in enabling privacy-enhancing quantization is not yet explored.

This work develops *Privacy-Utility Rate Adaptive Vector Quantization (PURAveQ)*, a learned vector quantization mechanism for remote inference that jointly achieves *multi-rate*, *privacy-aware*, and *task-based* compression. We leverage the Vector Quantization Variational Autoencoder (VQ-VAE) architecture [17] to learn task-oriented vector codebooks jointly with the encoder and inference modules, enabling end-to-end optimization for remote inference. We introduce a novel training procedure that incorporates: (i) the MI neural estimator (MINE) framework [30] to explicitly optimize the tradeoff between task utility and privacy leakage by estimating MI between latent representations and private attributes; and (ii) a progressive learning strategy that enforces the learned codebook to form a hierarchy of nested sub-codebooks, thereby supporting multiple quantization rates within a single unified model. Our numerical results demonstrate that, in settings where the MI between features and private attributes can be analytically computed, PURAveQ effectively mitigates the MI leakage related to private variables while maintaining notable utility information. Moreover, experiments on handwritten digit recognition confirm that the PURAveQ successfully obscures sensitive attributes while preserving reliable remote inference performance across varying bit rates.

II. SYSTEM MODEL AND PRELIMINARIES

A. System Model

Remote Inference: We consider a remote inference setting with a single sensing device and a remote user. At each time block t , the sensing device captures a sample $\mathbf{x}_t$, which is informative on both the desired utility, denoted $\mathcal{U}(\mathbf{x}_t)$, and on sensitive information, denoted $\mathcal{S}(\mathbf{x}_t)$. The sensing device communicates with the inferring device over a rate-limited channel modeled as a bit-pipeline with time-dependent capacity C_t holding $C_t \geq C_{\min} > 0$. The sensing device uses an encoder $\mathcal{E}(\cdot)$ to map $\mathbf{x}_t$ into a B_t bits representation $\mathbf{q}_t$ that is conveyed to the remote user at communication latency of $\tau_t = \frac{B_t}{C_t}$.

Problem Formulation: Our goal is to design the quantization mechanism $\mathcal{E}(\cdot)$ for private remote inference. The resulting $\mathbf{q}_t = \mathcal{E}(\mathbf{x}_t)$ is evaluated via the following measures:

M1 Utility, measured using the conditional entropy

$$M_U(\mathbf{x}_t; \mathbf{q}_t) = H(\mathcal{U}(\mathbf{x}_t) | \mathbf{q}_t). \quad (1)$$

M2 Privacy, measured using the *negative* conditional entropy

$$M_S(\mathbf{x}_t; \mathbf{q}_t) = H(\mathcal{S}(\mathbf{x}_t) | \mathbf{q}_t). \quad (2)$$

M3 Latency in transmitting the compressed representation, constrained by a maximum allowable delay $\tau_{\max} > \frac{1}{C_{\min}}$.

Measures *M1-M2* are often employed in information-theoretic privacy-utility analysis [29], while *M3* is a common constraint in remote inference [18], [20]. We do not require knowledge of the distribution of $\mathbf{x}_t$. Instead, for design purposes, we assume access to labeled data of the form $\mathcal{D} = \{\mathbf{x}_i, \mathcal{S}(\mathbf{x}_i), \mathcal{U}(\mathbf{x}_i)\}_{i=1}^{|\mathcal{D}|}$.

B. Preliminaries

Vector Quantization: Our methodology relies on data-driven vector quantizers. A vector quantizer with input dimension d is defined by a finite codebook $\mathcal{Q} \subset \mathbb{R}^d$, such that each input $\mathbf{x} \in \mathbb{R}^d$ is mapped into a B -bit representation $Q(\mathbf{x}) \in \mathcal{Q}$, where $B = \log_2 |\mathcal{Q}|$. This is typically carried out by selecting the closest codeword in $\mathcal{Q}$ [34, Ch. 23]. When $d > 1$, this operation is referred to as *vector quantization*, as opposed to *scalar quantization* for $d = 1$. Classical rate–distortion theory establishes that, under suitable statistical assumptions, vector quantization can achieve lower distortion than scalar quantization for the same bit budget [34]. The codebook $\mathcal{Q}$ can be optimized using multiple realizations of $\mathbf{x}$; a classic method for this task is the Linde–Buzo–Gray (LBG) algorithm [35], which iteratively selects $|\mathcal{Q}|$ codewords as the centroids of clustered data samples.

Data-Driven Vector Quantizers: DNNs were shown to learn multivariate quantized representations from data. A representative framework is the VQ-VAE [17] where an input $\mathbf{x}$ is processed by a DNN encoder into a latent representation $\mathbf{x}^e = f_e(\mathbf{x})$, typically a compact feature vector capturing the informative content of the input. The latent vector $\mathbf{x}^e$ is then decomposed into M sub-vectors of size d , denoted $\{\mathbf{x}_m^e\}_{m=1}^M$, and each sub-vector is represented by the closest codeword in $\mathcal{Q}$, i.e.,

$$\mathbf{q}^m = \arg \min_{\mathbf{q} \in \mathcal{Q}} \|\mathbf{x}_m^e - \mathbf{q}\|_2. \quad (3)$$

The quantized latent representation is thus $\mathbf{q} = Q_{\mathcal{Q}}(\mathbf{x}^e) = [\mathbf{q}^1, \dots, \mathbf{q}^M] \in \mathcal{Q}^M$, conveyed using $M \cdot B$ bits. A DNN decoder maps $\mathbf{q}$ into the reconstructed output $\hat{\mathbf{x}} = f_d(\mathbf{q})$.

To jointly train the encoder and decoder while learning the codebook, the VQ-VAE employs the following three-term loss:

$$\mathcal{L}^{\text{VQ}}(\mathbf{x}) = \mathcal{L}(\mathbf{x}; \hat{\mathbf{x}}) + \|\text{sg}(\mathbf{x}^e) - \mathbf{q}\|_2^2 + \beta \|\mathbf{x}^e - \text{sg}(\mathbf{q})\|_2^2, \quad (4)$$

where $\text{sg}(\cdot)$ denotes stop-gradient, $\mathcal{L}(\mathbf{x}; \hat{\mathbf{x}})$ is a reconstruction loss, and $\beta > 0$ is a hyperparameter. This loss balances codebook alignment with reconstruction accuracy [17]. Although several variants of this loss were proposed to enhance codebook utilization [36], (4) remains the prevailing VQ-VAE training loss [37].

MI and MINE: The MI between two random variables $\mathbf{x}$ and $\mathbf{y}$ quantifies their statistical dependence and measures how much

knowing one reduces uncertainty about the other. Formally, MI is defined as [6] $I(\mathbf{x}; \mathbf{y}) = \mathbb{E} \left[\log \frac{P(\mathbf{x}, \mathbf{y})}{P(\mathbf{x})P(\mathbf{y})} \right]$, where $P(\mathbf{x}, \mathbf{y})$ is the joint distribution, and $P(\mathbf{x})$, $P(\mathbf{y})$ are the marginals.

While the MI expression is well-defined, computing it analytically is generally intractable for high-dimensional or unknown distributions. A data-driven approach for MI estimation is the MINE framework [30], which leverages the Donsker–Varadhan representation [34, Ch. 3.3] to compute a trainable lower bound on the MI. Specifically, for a family of functions $T_\phi(\cdot)$ parameterized by a DNN with learnable weights ϕ , MINE estimates

$$\hat{I}_\phi(\mathbf{x}; \mathbf{y}) = \mathbb{E}[T_\phi(\mathbf{x}, \mathbf{y})] - \log \mathbb{E}_{P(\mathbf{x})P(\mathbf{y})}[e^{T_\phi(\mathbf{x}, \mathbf{y})}], \quad (5)$$

which satisfies $I(\mathbf{x}; \mathbf{y}) \geq \hat{I}_\phi(\mathbf{x}; \mathbf{y})$. By maximizing (5) over ϕ , MINE provides a differentiable data-driven estimate of MI.

III. PURAVEQ

Here, we introduce PURAVEQ, which is tailored for low-latency (*M3*) privacy-preserving (*M2*) and task-oriented (*M1*) remote inference over time-varying channels. Our design is based on the VQ-VAE, leveraging the combination of a DNN encoder and a learned codebook to obtain task-oriented vector quantized representations. As the VQ-VAE is static, supporting a single rate, and is not suitable for privacy preservation, PURAVEQ augments VQ-VAE-based quantization in the following aspects:

- A1 Multi-rate architecture* with a single learned codebook that subsumes multi-resolution sub-codes without retraining.
- A2 Privacy-utility-oriented training* that employs MINE-based privacy-utility regularization while leveraging progressive learning to construct nested sub-codebooks.

We next detail PURAVEQ, starting with its architecture (*A1*), followed by its training procedure (*A2*) and a discussion.

A. Architecture

Our design builds upon the VQ-VAE framework, implementing $\mathcal{E}$ using a DNN-based encoder f_e and a learned multivariate codebook $\mathcal{Q}$. Unlike conventional end-to-end methods, we focus on learning the quantization mechanism itself and not on specifying the decoder used by the inferring device. While one can optionally train a dedicated DNN decoder for downstream inference, our formulation is agnostic to its structure, allowing the quantizer to be integrated with various inference architectures.

Multi-Rate Codebook: We construct a VQ-VAE-based quantizer with a single codebook that supports multiple bit rates. The codebook is built iteratively, where each quantization level l expands upon its predecessor, effectively doubling its size. Formally, for quantization levels $l = 1, 2, \dots, \log_2 P$, we begin with a 1-bit codebook $\mathcal{Q}_1$, which is successively expanded to form higher-resolution codebooks $\mathcal{Q}_2, \mathcal{Q}_3, \dots$, according to

$$\mathcal{Q}_1 \subset \mathcal{Q}_2 \subset \dots \subset \mathcal{Q}_{\log_2 P}. \quad (6)$$

In this hierarchy, the first two codewords originate from $\mathcal{Q}_1$, the first four from $\mathcal{Q}_2$, and so forth. This allows a single codebook $\mathcal{Q}$ to be employed for multiple resolutions.

Rate Adaptation: To determine the appropriate quantization level for each transmitted sample, the sensing device selects the maximal level l_t satisfying the latency constraint:

$$l_t = \max \left\{ l \in \{1, 2, \dots, \log_2 P\} \mid \frac{l}{M \cdot C_t} \leq \tau_{\max} \right\}. \quad (7)$$

The encoder $\mathcal{E}(\cdot)$ and the full codebook $\mathcal{Q}$ are deployed at the sensing device, which dynamically selects the sub-codebook $\mathcal{Q}_{l_t}$ via (7). Accordingly, the quantized features at time t are set to

$$\mathbf{q}_t = \mathcal{E}(\mathbf{x}_t) = \mathcal{Q}_{\mathcal{Q}_{l_t}}(\mathcal{E}(\mathbf{x}_t)). \quad (8)$$

B. Training

We leverage the data $\mathcal{D}$ to learn a single VQ-VAE encoder $\mathcal{E}$ and codebook $\mathcal{Q}$ such that its quantized representations are useful in the utility-privacy sense **M1-M2** for bit-rate B_t that meets the required latency **M3**. We train PURAVeQ in three stages:

Stage 1: VQ-VAE Encoder Pre-Training: We first train solely the VQ-VAE encoder DNN $\mathcal{E}$ for non-quantized privacy-utility representations. To that aim, we leverage the fact that the loss measures in (1) and (2) can be respectively written as

$$M_U(\mathbf{x}; f_e(\mathbf{x})) = -H(U(\mathbf{x})) + I(U(\mathbf{x}); f_e(\mathbf{x})), \quad (9a)$$

$$M_S(\mathbf{x}; f_e(\mathbf{x})) = H(S(\mathbf{x})) - I(S(\mathbf{x}); f_e(\mathbf{x})). \quad (9b)$$

The marginal entropies in (9) are invariant of f_e , while the MIs can be estimated using MINE. We thus formulate a training loss

$$\mathcal{L}_D^{S1}(f_e) = \hat{I}(\{f_e(\mathbf{x}_i), U(\mathbf{x}_i)\}) - \lambda \cdot \hat{I}(\{f_e(\mathbf{x}_i), S(\mathbf{x}_i)\}). \quad (10)$$

In (10), $\lambda > 0$ balances the privacy-utility tradeoff, while $\hat{I}$ is an empirical estimate of (5), i.e., for samples $\bar{\mathcal{D}} = \{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^{|\bar{\mathcal{D}}|}$,

$$\hat{I}(\bar{\mathcal{D}}) = \max_{\phi} \frac{1}{|\bar{\mathcal{D}}|} \sum_{i=1}^{|\bar{\mathcal{D}}|} T_{\phi}(\mathbf{x}_i, \mathbf{y}_i) - \log \frac{1}{|\bar{\mathcal{D}}|^2} \sum_{i=1}^{|\bar{\mathcal{D}}|} \sum_{j=1}^{|\bar{\mathcal{D}}|} e^{T_{\phi}(\mathbf{x}_i, \mathbf{y}_j)},$$

which we approximate using MINE with T_{ϕ} being a DNN.

Stage 2: Codebook Initialization: After pre-training, the quantization codebook $\mathcal{Q}$ is defined with P codebook vectors, denoted $\{e_i\}_{i=1}^P$, each with embedding dimension d . To initialize these vectors, an unlabeled dataset $\mathcal{D}_Q$ is constructed by passing each $\mathbf{x}_i$ through the trained encoder to obtain $\mathbf{x}_i^e = f_e(\mathbf{x}_i)$, which is then partitioned into M subvectors $\mathbf{x}_{i,m}^e$ of size d . The LBG algorithm is applied to minimize quantization distortion:

$$\mathcal{L}_{\mathcal{D}_Q}^{S2}(\{e_k\}_{k=1}^P) = \frac{1}{|\mathcal{D}_Q|} \sum_{\mathbf{x}_i^e \in \mathcal{D}_Q} \min_{k=1, \dots, P} \|\mathbf{x}_{i,m}^e - e_k\|_2^2. \quad (11)$$

Stage 3: Joint Adaptation: The final stage jointly adapts the encoder f_e and the codebook $\mathcal{Q}$ starting from their values set in Stages 1-2, respectively. To support multiple rates, we adopt a form of progressive learning, where for each quantization level $l = 1, \dots, \log_2 P$, the codebook $\mathcal{Q}_l$ is constructed by expanding upon the previous level $\mathcal{Q}_{l-1}$, inheriting previously learned vectors and adding new vectors from the LBG initialization. The total loss at each level combines MINE-aided privacy-utility with VQ-VAE loss and rate adaptivity terms, and is given by

$$\begin{aligned} \mathcal{L}_D^{(l)}(f_e, \mathcal{Q}_l) = & \sum_{j=1}^l \left(\lambda \cdot \hat{I}(\{\mathbf{q}_i^{(j)}, S(\mathbf{x}_i)\}) - \hat{I}(\{\mathbf{q}_i^{(j)}, U(\mathbf{x}_i)\}) \right) \\ & + \frac{1}{|\mathcal{D}|} \sum_{i=1}^{|\mathcal{D}|} \left(\left\| \text{sg}(\mathbf{x}_i^e) - \mathbf{q}_i^{(j)} \right\|_2^2 + \beta_j \left\| \mathbf{x}_i^e - \text{sg}(\mathbf{q}_i^{(j)}) \right\|_2^2 \right) \\ & + \eta_l \sum_{k=1}^{2^{l-1}} \left\| e_k^{(l)} - e_k^{(l-1)} \right\|_2^2, \end{aligned} \quad (12)$$

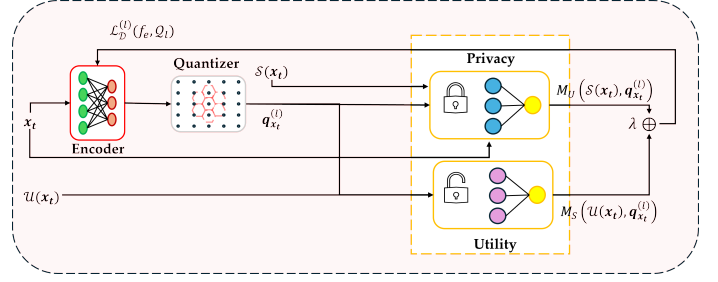


Fig. 1: Schematic illustration of training Stage 3

where $\mathbf{x}_i^e = f_e(\mathbf{x}_i)$ is the encoder output; $\mathbf{q}_i^{(j)} = \mathcal{Q}_j(\mathbf{x}_i^e)$ is the quantized representation of the encoder output using the first 2^j codebook vectors; and $\{\beta_l, \eta_l\}$ are hyperparameters. The first term in (12) encourages enhanced privacy-utility tradeoff of the quantized representation (see Fig. 1); the second term inherits the VQ-VAE codebook alignment and commitment terms; while the final term limits deviation of codewords optimized for lower rates. While (12) is formulated for a given resolution, the overall training scheme progressively updates f_e and $\mathcal{Q}$ from $l = 1$ to $l = \log_2 P$. The overall procedure is summarized as Algorithm 1.

Algorithm 1: PURAVeQ Training Procedure

Init: Init DNN f_e ; Set $\mathcal{Q}_0 = \emptyset$
Input: Dataset $\mathcal{D}$; Codebook size limit P ;
 Loss hyperparameters: $\lambda, \{\eta_l\}, \{\beta_j\}$
Stage 1: VQ-VAE encoder pretraining;
 1 Train f_e to maximize $\mathcal{L}_D^{S1}(f_e)$ (10);
Stage 2: Codebook Initialization;
 2 Create dataset $\mathcal{D}_Q = \{\mathbf{x}_i^e\}_{i=1}^{|\mathcal{D}|}$;
 3 Use LBG on $\mathcal{D}_Q$ to initialize codewords $\{e_k^{\text{LBG}}\}_{k=1}^P$;
Stage 3: Joint Adaptation;
 4 **for** $l = 1, \dots, \log_2 P$ **do**
 5 Initialize $\mathcal{Q}_l = \mathcal{Q}_{l-1} \cup \{e_j^{\text{LBG}}\}_{j=2^{l-1}+1}^P$;
 6 Train $(f_e, \mathcal{Q}_l)$ to minimize $\mathcal{L}_D^{(l)}(f_e, \mathcal{Q}_l)$ (12);
Output: Trained encoder f_e and codebook $\mathcal{Q}_{\log_2 P}$

C. Discussion

PURAVeQ meets the design requirements set forth in Subsection II-A. By incorporating MINE-based privacy-utility regularization, PURAVeQ learns encoder mappings and codebooks that maximize the utility MI while suppressing private information, thus quantizing to preserve task-relevant features and obscure sensitive ones. Through its codebook construction, PURAVeQ supports multiple bit rates without retraining or structural modification. Once trained, PURAVeQ operates as a lightweight VQ-VAE encoder that maps each input sample to its nearest codeword in the appropriate sub-codebook. While the current design assumes access to the current channel rate to set its bit level, it can be potentially extended to support adaptive rate and set the bit level on-the-fly. Also, while we focus on compression with channels modeled as noiseless bit-pipes, the framework can be extended toward privacy-utility-aware deep joint source-channel coding [38]. Such extensions are left for future work.

IV. NUMERICAL EXPERIMENTS

Scenarios: We evaluate PURAVeQ in three scenarios: (i) a synthetic data where MI can be evaluated analytically; and

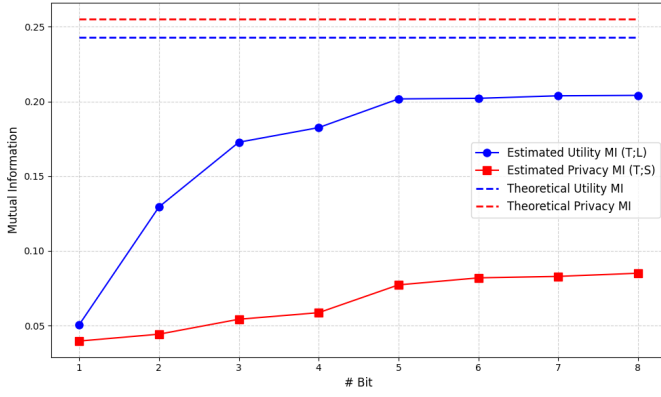


Fig. 2: MI versus bit-rate, synthetic setup

image classification with (ii) MNIST and (iii) CIFAR-10. In the synthetic case, we set $\mathcal{U}(\mathbf{x})$ and $\mathcal{S}(\mathbf{x})$ to be two-dimensional mutually independent Gaussian vectors with distinct means, where $\mathbf{x}$ is their sum corrupted by zero-mean Gaussian noise. For MNIST, the utility is whether the digit is odd or even, while the private information is defined as whether the digit is larger than 4. For CIFAR, the utility is set to the superclass (i.e., animal or vehicle), while the private information is the actual label¹.

Architecture: The DNN f_e is implemented as a three-layer convolutional neural network (CNN) with a fully connected (FC) output layer. The codebook has $P = 256$ codewords with embedding dimension $d = 4$. To estimate MI from data, we implemented the MINE DNN T_ϕ using three FC layers.

Synthetic Data Results: In the synthetic case, the MIs $I(\mathbf{x}; \mathcal{U}(\mathbf{x}))$ and $I(\mathbf{x}; \mathcal{S}(\mathbf{x}))$ can be computed analytically due to the joint Gaussianity of the signals. Thus, this setup allows us to assess the reduction in each MI term (which is guaranteed by the data processing inequality [6]) due to PURAVeQ. The resulting MI values per quantization level l are reported in Fig. 2, where the MIs before PURAVeQ are computed analytically, and the MIs of the quantized values are estimated with MINE. We observe in Fig. 2 that while the input to PURAVeQ is more informative with respect to the private features, the quantized output notably changes this. Still, quantization induces some loss in the utility MI, which is highly coupled here with the private features, and the loss depends on how coarse the quantization is.

Image Classification Results: In non-synthetic datasets, we do not have an analytical expression for the MIs. Thus, to evaluate PURAVeQ, we use t-SNE visualization of the quantized representations. Specifically, in Fig. 3 we show the evolution of the t-SNE values for the quantized representation with $l = 3$ bits per $d = 4$ vector during training for MNIST, while marking the utility and privacy labels with distinct colors. We observe in Fig. 3 that the training procedure results in utility information being distinguishable in the quantized representation, while the private labels become obscured as training proceeds.

To demonstrate how our MI-guided multi-rate quantization reflects on the ability to recover the utility and private information, we train dedicated classifiers based on the output of PURAVeQ for both $\mathcal{U}(\mathbf{x})$ and $\mathcal{S}(\mathbf{x})$. Each classifier is a two-layered FC network, trained with the cross-entropy (CE) loss to classify its corresponding label from $\mathcal{E}(\mathbf{x})$. We consider both

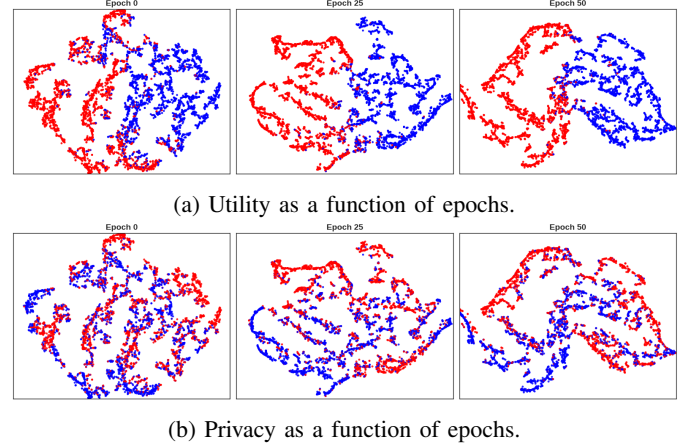


Fig. 3: t-SNE visualization, MNIST

Method	Comp. Rate (%)	MNIST		CIFAR10	
		Util.	Priv.	Util.	Priv.
PURAVeQ	0.7	71.7	55.9	74.0	24.0
	1.5	94.0	58.0	76.5	25.6
	2.3	94.2	58.0	77.6	27.6
	3.1	94.2	58.0	79.1	29.7
	3.9	94.4	58.2	80.5	31.5
	4.6	94.4	58.2	81.3	31.6
	5.4	94.4	58.2	81.5	32.0
Unquantized, MINE loss		92.8	56.4	79.9	27.9
Unquantized, CE loss		99.8	98.9	93.3	63.6
Random Guess		50.0	50.0	60.0	10.0

TABLE I: Utility/privacy accuracy (%), CIFAR10 and MNIST the case where $\mathcal{E}(\cdot)$ is PURAVeQ, trained once and evaluated with different resolutions, as well as without quantization (*Unquantized*), i.e., $\mathcal{E} \equiv f_e$, with full (32-bit) precision representation. For the unquantized encoder, we considered both privacy-utility aware training using the MINE-aided loss of (10), as well as when the encoder is trained end-to-end along with the utility and private features classifiers via the CE loss. The resulting utility/privacy accuracy for both MNIST and CIFAR-10 are reported in Table I, along with the achieved compression rate (with 100% representing the unquantized representations).

We observe in Table I that a single trained PURAVeQ model can provide representations at various different compression ratios, while simultaneously enabling accurate recovery of the utility information and obscuring the ability to faithfully extract the private information. Specifically, when compressing to merely 3% of the unquantized representation, PURAVeQ exhibits near end-to-end utility accuracy and near random-guess privacy extraction accuracy for MNIST, with larger gaps observed for CIFAR, where there is more entanglement between the utility (super-class) and private information (class).

V. CONCLUSIONS

We introduced PURAVeQ, a task-based, privacy-preserving, and rate-adaptive vector quantizer remote inference over time-varying communication channels. By combining a nested codebook VQ-VAE with MINE-based optimization, PURAVeQ simultaneously preserves task-relevant information while obscuring private information and supporting different compression rates.

¹The source code can be found at <https://github.com/eyalfish/PURAVeQ>

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